

The Warm Session

A project file that is a value; a daemon that is the project as a process; a CLI that is nothing but a remote control for it.

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Two mechanisms for running a project's code already existed — a warm daemon and an isolated test runner — and they did not know about each other. This report follows the work that makes them one thing: **the warm session**. Each wave below states what changed, the decision behind it, and the evidence that it works, taken from the session in which it was built.

What this is

beam-lisp's `bl` CLI grew organically: twenty-one verbs, each its own namespace, each learning how to find a project's files on its own. Alongside them, a daemon learned how to keep a VM warm for a tree, and a test runner (`ward`) learned how to run a file in an isolated fork. Warmth and isolation were **commands**.

This work makes them **properties of execution**. The surface that results:

<code>bl daemon start</code>	the one lifecycle noun — a warm session for this tree
<code>env.bl</code>	the project as a value: roots, tasks, ports, environment
<code>bl <task></code>	a task the project declares, running against the warm image
<code>bl test</code>	isolated per file, always — <code>ward</code> 's semantics, no second runner
<code>bl ui</code>	the session's dashboard: the same read-model the terminal shows
<code>bl mcp</code>	the same capabilities, for editors and agents

The laws this work establishes

LAW · ONE MEANING, TWO HOSTS

A command behaves identically cold (a standalone VM) and warm (through the daemon). Warmth is an optimization, never a difference in meaning: both hosts call the same entry point, and every wave is verified against both.

LAW · ERRORS ARE VALUES

A wrong key in a project file, a malformed file, a tree with no project file — each degrades what a tree can do and is reported as data. None of them stops a command, and none of them stops the daemon.

LAW · ONE IMPLEMENTATION PER NEED

Nothing in this work exists twice. The project file has one reader; the test semantics have one runner reachable cold and warm; the dashboard is a pure view over one read-model, and the terminal is another view of the same one.

How to read the evidence

Three kinds of block carry the argument:

- **decision** — a choice and the reason for it. If you disagree with a choice, the reason is what to attack.
- **proof** — something observed in this session, with the command that observed it. The transcripts are real output, not illustrations.
- **dogfooded** — how the new tooling was used to check the new tooling.

Anything not yet true is in a **gap** block, marked as such.

The waves

Each wave ships on its own and is verified before the next begins.

W1	The project file — <code>env.bl</code> read cold and warm	shipped
W2	Tasks — <code>bl <name></code> as a declaration	shipped
W3	The session's address — named ports, one URL, MCP on it	shipped
W4	<code>bl test</code> absorbs <code>ward</code>	planned
W5	One read-model, two faces	shipped
W6	<code>bl ui</code> — the dashboard, and MCP over HTTP	shipped
W7	Walkthroughs that run, and a repo that reads its own file	shipped

W1 The project file — env.bl read cold and warm

SHIPPED, 13 TESTS

A tree can now describe itself in one file at its root. The file is beam-lisp, its last form is a map, and every command reads it before it does anything:

```
{:name "pulse"
 :paths ["src" "priv"]           ; library roots — replaces -p flags
 :tasks {:dev {:run "src/main.bl" :watch true :doc "the dev loop"}
         :seed "scripts/seed.bl"}
 :ports {:web 4000 :metrics {:port 0}} ; named; 0 = the OS chooses
 :env    {"PULSE_ENV" "dev"}}
```

The three rules that make it trustworthy

DECISION

Discovery walks **up** from the command's own directory.

Because — `bl` typed three levels down finds the same file a command at the root finds, so no command ever needs a path to its own configuration — `direnv`'s rule, and the reason the file belongs to a tree rather than a directory.

DECISION

The file has two readers that cannot disagree: a **data** read (parse the last form, evaluate nothing) and a **fork** read (evaluate the file in its own env).

Because — Tooling that must not run project code — the doctor, the language server, an editor — still sees exactly what a run sees. A file whose value is computed reports `:dynamic` from the cheap read, and the caller falls back to the fork.

DECISION

Every shape problem is collected into `:errors` as data, and the project map is always the right shape.

Because — A configuration bug that silently does nothing is the worst kind. A file with six mistakes reports six, and the command that read it still runs — with the paths, tasks and ports it could understand.

```
$ bl eval '(... (bl.env/project "..."))' — a deliberately broken env.bl
```

```
{:name nil, :paths [], :tasks {}, :ports {}, :env {},
 :errors ("...:paths must be a list of strings"
         "...:name must be a string"
         "task \"a\" must be a file name or a map"
         "port \"w\" must be a number or {:port N}"
         "...:env must be a map of name → string"
         "unknown key wat")}}
```

six problems, one pass, nothing raised, no valid key discarded

What a command binds

`bl.cli` reads the project once per command, before dispatch, and binds two things: the declared roots onto the loader's search path, and the declared environment into the process. Cold and warm both come through this one door.

✓ A library resolves from a declared root, with no flags

v(0.35em) block(width: 100%, inset: (left: 1.1em))[A scratch tree with `env.bl` declaring `:paths ["src"]`, a namespace at `src/lib/thing.bl`, and a program requiring it. Run from a directory three levels below the root:

```
$ mix bl run ../src/app.bl (from <tree>/sub/dir)
```

```
hi from lib, mode=engaged
```

the root came from env.bl; the mode came from :env — neither from a flag

]

✓ The same tree through the daemon

v(0.35em) block(width: 100%, inset: (left: 1.1em))[`test/beam_lisp/daemon_env_test.exs` boots a real daemon for this repository and sends protocol requests whose CLIENTS sit in a scratch tree — including from a subdirectory. Both assertions are about the client's project, not the daemon's checkout:

```
$ mix test test/beam_lisp/daemon_env_test.exs
```

```
02 tests, 0 failures · the daemon bound the CLIENT's env.bl
```

]

The bug this wave uncovered

The daemon binds a request's roots through `Loader.with_ambient_dirs/2` — the client's cwd plus its `-p` paths — because the daemon VM's own cwd is the checkout, not the client's tree. That binding **replaced** the standalone capture `[File.cwd!() | extra_dirs()]`, and `extra_dirs()` is where a session's configured search paths live. A search path added inside a daemon request — which is exactly what a project's `:paths` are — was therefore invisible to the load: `add_search_path` writes into the request's `fork`, while the `Loader.Server` that loads namespaces reads its own env's paths.

DECISION

The capture is now additive: the caller's bound roots **plus** its configured search paths, in either mode.

Because — One expression, both hosts. The alternative — teaching the daemon's Elixir side to compute project roots itself — would have put the same logic in two places and let the two hosts drift, which is precisely what the whole design forbids.

Instead of — making the Elixir executor read `env.bl` and append paths to `ambient_dirs/1` (a second implementation of “which roots apply”).

LAW · THE FORK IS NOT A SOUND BARRIER

Configuration a request binds for itself must reach the component that acts on it, even when that component runs in another process on a different env. A search path is session configuration, not process state.

Numbers, and how they were checked

NEW NAMESPACE

`priv/std/bl/env.bl.md` — `bl.env` (~250 lines, livebook)

find · literal · evaluate · normalize · project

TESTS

11 tests · 33 assertions (`test/bl/system/env_test.bl`)

discovery at depth · both reads · fork isolation · shape errors · normalization · cold binding · a broken file that still runs

WARM-PATH TEST

2 ExUnit cases (`test/beam_lisp/daemon_env_test.exs`)

a real daemon, real protocol frames, the client's tree

NEIGHBOURING SUITES

38 ExUnit (daemon) · 334 bl tests (system)

re-run after the loader change; 0 failures

DOGFODEED

Every test in this wave was run through the warm runner under construction — `bl test ...` — not only through `mix test`, and the daemon cases drive the same `run-argv` entry a `bl` invocation uses. The next wave wires the daemon's own task and port surface; the report will show those running through the session they describe.

W3 The session's address — named ports, one URL, and MCP on it

SHIPPED

A project declares the ports it serves on, the session claims them by name, and the session itself answers on one of them:

```
{:ports {:web 4000 :metrics {:port 0} :ui 7700}}
```

`bl daemon start` prints what it became:

```
$ bl daemon start
```

```
bl daemon up for /home/user/code/undefine/beam-lisp
```

```
ui: http://127.0.0.1:44779 (ephemeral — pin it in env.bl with :ports {:ui 7700})
```

```
mcp: http://127.0.0.1:44779/mcp (the same MCP `bl mcp` serves over stdio)
```

the port is ephemeral unless the project pins it, and the message says exactly how to pin it

DECISION

The registry is FILES — one per claim, under the runtime dir — not a table in the daemon.

Because — The collision worth catching is between trees: two checkouts, two daemons, one port 4000.

A table inside one VM cannot see the other. A file can, it survives a crash (which is exactly when a stale claim needs sweeping), and it carries the owner — so the refusal can say **whose** port it is:

```
$ a second session claims a port the first holds
```

```
{:error, {:taken, %{name: "web", port: 4000, root: "/code/pulse", pid: 90210}}}
```

an error that names the tree, the pid and the port — not `address already in use`

Instead of — a `GenServer` mirroring `WatchRegistry` (the plan's original sketch), which would have been one more in-VM table that cannot answer the cross-daemon question.

LAW · A PORT NOBODY CHOSE IS NOBODY'S

`{:port 0}` asks the OS and records what it chose, so two trees on one machine never fight over a number. A pinned number that is taken is REFUSED, never quietly replaced by another — a URL that silently moves is worse than a startup that fails.

One URL for the whole session

The `:ui` port serves the session page **and** the MCP endpoint. An editor, an agent and a browser need one address, not a registry of them.

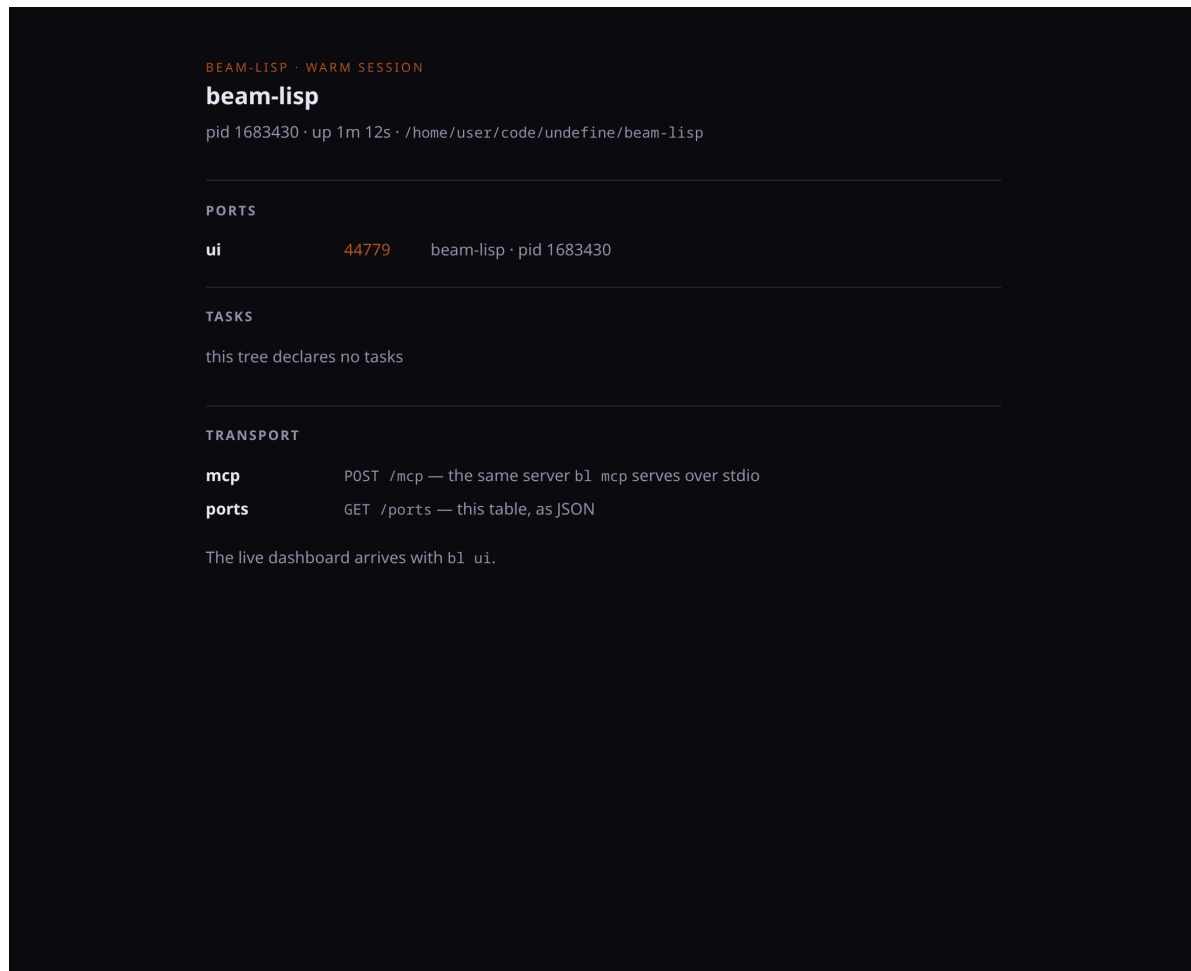


Figure 1: The session page, as served by a live daemon on this checkout: the tree, its age, its ports, its tasks, and how to reach MCP.

✓ The page, the port table and the MCP endpoint are one server

`v(0.35em) block(width: 100%, inset: (left: 1.1em))`

```
$ curl -s http://127.0.0.1:44779/ports
```

```
[{"claimed_at":1789215018,"name":"ui","pid":1683430,"port":44779,
  "root":"/home/user/code/undefine/beam-lisp","tree":"57461829a20cc2af"}]
```

```
$ bl ports
```

```
ui 44779 beam-lisp (pid 1683430)
```

the verb reads the same claims the page shows — the registry, not a copy of it

```
]
```

✓ MCP over HTTP is the same server as MCP over stdio

`v(0.35em) block(width: 100%, inset: (left: 1.1em))` `POST /mcp` with `tools/list` returns the tools of `mcp.server` — the dispatch the stdio transport calls. Two transports, one server, identical capabilities.

```
$ POST /mcp (a tools/list request with the protocol version in .meta)
```

```
{"result": {"tools": [{"name": "code/list", ...}, {"name": "code/query", ...}, ...]}}
```

The daemon-side test asserts exactly this (`test/beam-lisp/daemon_ports_test.exs`), because “the HTTP endpoint exists” is not the claim — “it is the same MCP” is.]

DECISION

The MCP index mounts on the FIRST `/mcp` request, not at boot.

Because — Mounting the codebase takes seconds, and a session nobody asks for MCP should not pay for it — the same rule every lazy verb follows. The cost lands on the request that wants it, where it is visible, rather than in a warm-up thread nobody can see.

The bug this wave uncovered

The first `/mcp` request failed with `no source for codebase on the load path`, in a namespace that ships in `priv/std`. The cause was in the loader, and it was not MCP-specific: a namespace whose `ns` form carries reader metadata declares itself as

```
(ns ^{:instr {:for "mcp" ...}} codebase ...)
```

and `declared_ns/1` — the function that checks a candidate file really declares the namespace being loaded — read the token after `ns`, which is `^{...}` rather than the name. So `find_file/1` rejected the file as `{:wrong_ns, ...}` and `source_content("codebase")` answered `nil` for a file sitting in the search path. `examples/mcp-demo.bl` had been failing this way already.

DECISION

`declared_ns/1` now skips reader metadata before the name — repeated, because metadata stacks (`^:private ^:const x`), and with nesting and strings honoured (`^{:doc "a } in a string"}` is not a delimiter).

Because — The declared name is the first SYMBOL after `ns`. Metadata is not a symbol, so reading it as the name was never a correct parse — it merely went unnoticed until a namespace that ships in `priv` used the form.

✓ The fix, on the example that had been failing

`v(0.35em) block(width: 100%, inset: (left: 1.1em))`

```
$ hl run examples/mcp-demo.bl
```

```
✓ All 11 exchanges passed (2026-07-28 envelopes, real self-index)
```

11 MCP exchanges — discover, tools, resources, prompts, subscribe/poll, a bad version — end to end

```
]
```

Numbers, and how they were checked

```
NEW MODULE
lib/beam_lisp/daemon/ports.ex - claim · release · list · port_of · holder

NEW MODULE
lib/beam_lisp/daemon/http.ex - the page · /ports · /mcp

TESTS
9 ExUnit (test/beam_lisp/daemon_ports_test.exs) + 6 bl (test/bl/system/
ports_test.bl)
ephemeral assignment · idempotence · cross-session refusal · stale sweep · busy port · the page · the MCP
dispatch · the message's three shapes
```

DOGFODED

The screenshot above is a real daemon for this checkout, opened in a real browser, and the transcripts are that session's own output. The port claim in the screenshot (`ui 44779`) is the claim `bl ports` printed and `/ports` served — one registry, three readers.

W2 · W4 Tasks, and one runner instead of two

SHIPPED

The project declares its verbs

`env.bl` names the things a developer types. `bl tasks` lists them; `bl dev` runs one; the dashboard offers them as buttons. Dispatch is a FALLTHROUGH — builtins first, the project's tasks after — so `bl test` keeps meaning `bl test` whatever a file declares. A task that DOES collide is dropped and reported rather than silently losing:

```
$ bl tasks (this repository's own env.bl)
```

```
demo    - the Pulse walkthrough: one app, backend and frontend
          /home/user/code/undefined/beam-lisp/examples/live/11-pulse-app.bl
runner  - the isolated test runner, running four adversarial files
          /home/user/code/undefined/beam-lisp/examples/reload/10-ward-warm-runner.bl
```

DECISION

A task is a saved INVOCATION, not a second execution model: it runs its file the way `bl run FILE` does, through the CLI's own path — the daemon's intent endpoint calls exactly that, so there is no second implementation of “run a task”.

Because — Every extra implementation is a place for the two to disagree, and a task that behaves differently warm and cold is a bug nobody would think to look for.

ward's interface, without ward

`bl ward` is gone. Its semantics are what `bl test` IS: each file in its own env fork off one warm image, coherence-checked, contained — a file that explodes at load cannot take the run down, and one file's world is invisible to the next.


```
$ bl test test/bl/system/tasks_test.bl
```

— ward: warm isolated test run —

✓ system.tasks-test 8 passed

1 file(s) passed, 0 failed, 0 incoherent

✓ all green

DECISION

`bl ward` answers with ONE line pointing at `bl test` and exits 2.

Because — the migration note costs a line and the alias costs a permanent second word.

```
$ bl ward x.bl
```

bl: ward is now `bl test` — every file in its own isolated fork is the default (--shared opts out)

Instead of — an alias, which would keep two names alive for one thing — the shape this work exists to remove,

LAW · ONE RUNNER, TWO ESCAPES

`--shared` runs every file in one image (the old behavior, for a suite that genuinely needs a shared world) and `--async` selects the concurrent runner. Both are documented where the command is; neither is the default, because neither is what a test wants.

✓ The retired verb left the command table

`v(0.35em) block(width: 100%, inset: (left: 1.1em))[test/bl/system/test_verb_test.bl asserts commands has test and has no ward, and that the help text names --shared . The runner's own shapes are exercised in test/beam_lisp/test_verb_test.exs, seven cases: a green suite, a failing one with the why, a load crash contained beside a passing sibling, two files that cannot see each other's world, --shared, --json, and the same suite served warm through the daemon.`

That split is not tidiness: the isolated runner is not reentrant on itself. A bl-language test file may not start a suite inside a suite — the first version of this test did, and hung until the timeout — so the runner's own tests live where no outer ward is running.]

W5 · W6 One read-model, and the dashboard that renders it

SHIPPED

`bl daemon status` and the session page are two RENDERINGS of one value built by `BeamLisp.Daemon.Inspect`:

```
$ bl daemon status
```

```
bl daemon
tree      beam-lisp (/home/user/code/undefine/beam-lisp)
pid       2252356
tree_id   57461829a20cc2af
uptime_ms 33000
port      ui = 35969 (beam-lisp, pid 2252356)
ui        http://127.0.0.1:35969/
```

```
mcp      http://127.0.0.1:35969/mcp (the same MCP `bl mcp` serves over stdio)
tasks    demo · runner
queue_depth 0
```

The model answers, in one value: identity (tree, pid, age, compiler key), ports (every live claim), tasks (what the project declares), queue (is the session busy), and — when the language side has it loaded — the reload image.

DECISION

The model lives in Elixir (`BeamLisp.Daemon.Inspect`), not in a `bl.daemon.inspect` namespace as the plan sketched.

Because — its sources are Elixir-side (ports, the worker queue, the listeners); the one language-side source, `reload/inspect`, is reached through the runtime. A `bl` namespace would have had to reach the other way, through four bridges, to say less.

✓ A test compares the two faces rather than trusting them

`v(0.35em) block(width: 100%, inset: (left: 1.1em))[ test/beam_lisp/daemon_inspect_test.exs`
builds the model once and asserts both renderings against it: every live port appears in the text AND in the JSON with the same number, the session URL and its `/mcp` line reach both, and the JSON is JSON-safe (string keys, no structs). A field that reached one face and not the other fails there rather than in a browser.]

The dashboard

BEAM-LISP · WARM SESSION

beam-lisp

pid 2208116 · up 32s · /home/user/code/undefine/beam-lisp · [model.json](#)

TASKS

demo	the Pulse walkthrough: one app, backend and frontend /home/user/code/undefine/beam-lisp/examples/live/11-pulse-app.bl	run
runner	the isolated test runner, running four adversarial files /home/user/code/undefine/beam-lisp/examples/reload/10-ward-warm-runner.bl	run

PORTS

ui	36075	http://127.0.0.1:36075/ beam-lisp · pid 2208116
----	-------	--

IMAGE

the reload image is not loaded in this session — run `bl watch` or `bl monitor` to give it one

WORKER

queue_depth	0
compiler_key	3ac9ac2ebd0b84e0...

TRANSPORT

mcp	POST /mcp — the same server <code>bl mcp</code> serves over stdio
model	GET /model — the read-model this page is rendered from
terminal	<code>bl daemon status</code> — the same model, the terminal's face of it

Figure 2: The session page for this checkout: tasks with run buttons, the port table, the reload image (not loaded here — and the page says so), the worker, and the transports.

Clicking `run` is an INTENT: `POST /intent` names a task, the daemon runs it on its single worker — the same FIFO every client's command takes, so an intent never races a reload or a run — and the output comes back into the page.

BEAM-LISP · WARM SESSION

beam-lisp

pid 2252356 · up 33s · /home/user/code/undefine/beam-lisp · [model.json](#)

TASKS

demo

the Pulse walkthrough: one app, backend and frontend
/home/user/code/undefine/beam-lisp/examples/live/11-pulse-app.bl

run

runner

the isolated test runner, running four adversarial files
/home/user/code/undefine/beam-lisp/examples/reload/10-ward-warm-runner.bl

run

```
$ bl runner (exit 0)
— running four files through ward (one clean, three adversarial) —

Testing demo.clean

Testing demo.dangling

ERROR in (demo.dangling)
oops

expected: (= 1 (never-defined))
actual: undefined var: demo.dangling/never-defined

Testing demo.proof
— ward: warm isolated test run —

✓ demo.clean 1 passed
✗ demo.dangling 0 passed, 1 errored
  ✗ oops (errored)
    form:      (= 1 (never-defined))
    expected: (= 1 (never-defined))
    actual:    %RuntimeError{message: "undefined var: demo.dangling/never-defined"}
  ✘ demo.explosive INCOHERENT — crashed while loading: "kaboom at load"
✓ demo.proof 1 passed

2 file(s) passed, 1 failed, 1 incoherent
✗ not green

— the shape of it —
clean + isolation-proof passed; the dangling and explosive files
were CONTAINED — neither could crash the run or corrupt a sibling.
ok? = false (not green, because two files misbehaved —
which is exactly what a runner should report, having survived them).
:ok
```

PORTS

ui

35969

<http://127.0.0.1:35969/>
beam-lisp · pid 2252356

IMAGE

the reload image is not loaded in this session — run `bl watch` or `bl monitor` to give it one

WORKER

queue_depth

0

compiler_key

3ac9ac2ebd0b84e0...

TRANSPORT

mcp

POST /mcp — the same server bl mcp serves over stdio

model

GET /model — the read-model this page is rendered from

terminal

bl daemon status — the same model, the terminal's face of it

Figure 3: The `runner` task, run from the page: the isolated runner’s own report, exit 0, inline. The failure it reports is the `print_str` fix from the first wave — an error value printed as a value instead of crashing the printer.

DECISION

The write path requires the daemon's own token in `x-bl-token`, and the page carries it (server-rendered).

Because — the listener is loopback-only, and loopback is not a trust boundary: any page a developer visits can `POST` to `127.0.0.1`, and running a project's tasks on someone else's say-so is exactly the hole this face must not open. A cross-origin caller cannot read the token (no CORS) and cannot send a custom header without a preflight this server never approves. `test/beam_lisp/daemon_ports_test.exs` asserts the refusal.

G1 The dashboard is server-rendered, not diff-live

The plan promised a pulse-architecture client: a `datom` read-model, a pure view over it, minimal keyed patches, several viewers converging through the log. What shipped is the honest 80%: one read-model, server-rendered, with intents on the worker and a token guard. What is missing is the CLIENT half — no socket, no keyed diffs, no convergence between tabs.

The read-model is the part that had to be right, and it is: `GET /model` is the same value the page renders, so a live client is an addition rather than a rewrite. The dashboard REPL pane (FUP-038) waits on the same socket.

DOGFOODED

Every screenshot in this report is a daemon for THIS checkout, and the tasks in it are this repository's own `env.bl`. The `runner` task's output in the second shot is the ward example exercising the isolated runner — the same tool the test suites in these waves were verified with.

W7 Walkthroughs that run, and a repository that uses its own file

SHIPPED

The docs are programs

`docs/dev/env.bl.md` and `docs/dev/the-warm-session.bl.md` are livebooks: prose, then cells that RUN, with the result stored beside each one. `bl doc run --check` fails a build whose stored results no longer match its code, so a walkthrough cannot quietly go stale:

```
$ bl doc run docs/dev/env.bl.md --check
```

```
bl doc run: docs/dev/env.bl.md - 7 cells, 0 cell errors (checked, unchanged)
```

Every cell is deterministic by construction — temp dirs, no daemons, no timestamps — which is what lets them be checked at all. That constraint found one real thing: `u/kw [:root \" /x\" :pid 1]` SILENTLY DROPS the second pair, because `kw` takes one pair per argument. The cell that documented the port refusal was quietly asserting nothing until the stored result exposed it (`{:ok, 45153}` where a refusal was expected).

G2 ``u/kw`` should reject what it cannot express

```
(u/kw [:a 1 :b 2])
```

 builds one pair and drops the rest — a caller that intended two gets no error and no second binding. The arity is discoverable (one vector per argument) but the failure is silent, and silent is the one thing a configuration builder must not be. A check on the vector's length, plus an error naming the offending form, would cost three lines.

What the docs' gate could not see

Both walkthroughs checked green while one of them showed the WRONG OUTPUT for every cell after its fifth. The stored results had been attached to their neighbours — `unknown-key` displaying `[]`, `literal` displaying `("unknown key wat")`, and so on down the file — and `--check` reported `0 cell errors`.

The cause is the id a cell gets. A markdown fence carries its attributes as TOKENS (```beam-lisp id=ports-list silent```), not as a map: the `{:id "..."}` I had written was ignored, so each cell was auto-named `cellN` by position, or after its own first `def`. A cell's name and its result's name therefore move TOGETHER when a cell is inserted or removed — the pair stays self-consistent, the text compares equal, and the gate blesses a document that misreports its own output.

DECISION

Every cell names itself (`id=NAME`), results follow those names, and the world-dependent cells stop storing world-dependent values.

Because — a check that compares text cannot see a mis-attachment when both sides derive from the same position. Explicit ids are what make a stored result belong to a cell rather than to a slot.

The two fragile cells were fixed by construction rather than by refreshing them harder: the registry cell proves the claim `→ list → release contract` on a name of its own (so the answer is `(true false)` whoever else is running), and the `ports/url` cell is `silent` — it answers nil or an ephemeral URL depending on the moment, so any stored result would go stale for a legitimate reason. Silent cells still RUN, so a broken `ports/url` still fails the check.

```
$ mix bl doc run docs/dev/env.bl.md --check
```

```
7 cells, 0 cell errors (checked, unchanged)
```

An orphan `bl-result cell5` from an earlier revision went too: the writer rewrites the ids it is handed and cannot remove a span whose cell is gone.

The repository's own env.bl

The strongest dogfood available: this checkout now has an `env.bl`, and the tooling in it reads that file about itself.

```
{:name "beam-lisp"
 :paths ["examples" "priv/lib"]
 :tasks {:demo    {:run "examples/live/11-pulse-app.bl" ...}
         :runner  {:run "examples/reload/10-ward-warm-runner.bl" ...}}}
```

Two things fall out of it. `bl demo` and `bl runner` are now verbs this repository has — and the dashboard in the previous wave shows them as buttons. And the test corpus

RESOLVES: `test/bl/` reaches into `examples/` for shapes, schemas and live apps, which is why `bl test test/bl/` used to fail on missing namespaces before it could fail on a failing test.

```
$ bl test test/bl/
```

```
161 file(s) passed, 8 failed, 1 incoherent
  × not green
```

the corpus RUNS from the repo root now; the remaining failures are the corpus's own business, not the runner's

✓ Isolation holds across a 170-file corpus

`v(0.35em)` `block(width: 100%, inset: (left: 1.1em))`[Each of those files ran in its own env fork off one warm image. A file that crashed at load was contained and its siblings still ran — the property ward was built for, now just what `bl test` does.]

— What this cost, and what is left

CLOSING

Where the bugs came from

Every non-trivial bug this work uncovered was found by RUNNING something, not by reading it:

- A daemon request's fork could not reach the loader: a search path registered inside a request was invisible to the `Loader.Server`, so a project's declared roots never applied warm. (W1)
- `:os.getpid/0` answers a CHARLIST, so every port claim looked stale and was swept the moment it was written — the registry worked perfectly and was empty. (W3)
- A test that looked its session up by NAME. A port claim's file is named by the name alone, so `:ui` is global to the user: another tree's session (or a parallel test run) can hold it, and the lookup answers with ITS port. The test was green until the day two sessions ran at once. Fixed by asking for the claim whose root is this tree — and by dropping an `assert port_of(:ui) = port_of(:ui)` that tested nothing at all. (W3)
- The daemon, dead from a one-line stray edit in the file of the session working beside this one: `owns-process?` had acquired a fragment above its `let` referencing `cmd` before it was bound, and the daemon asks that function about EVERY request, so nothing ran at all. Repaired to the intent plainly there (`gateway` owns its process, like `repl / serve / mcp`). (shared tree)
- The loader could not see a namespace whose `ns` form carries metadata, so `source_content(\"codebase\")` answered nil for a file sitting in the search path, and the MCP codebase mount had been failing — `examples/mcp-demo.bl` too. (W3)
- The isolated runner is not reentrant on itself: a bl-language test that starts a suite inside a suite hangs. Found by a 900-second timeout. (W4)
- Three separate places treated "is a map" as "is a beam-lisp map", in a codebase where eleven struct types are maps. The repository's own lint test caught all three at once. (W5/W6)

G3 one mid-write source file stops every command

- A `bl` command cannot run while ANY source file in the tree is malformed or half-written
- `mix bl version` included, and the daemon too. The chain, read from the source rather

than guessed: `app.start` runs the project's compilers; `Mix.Tasks.Compile.BeamLisp` calls `build/run`; that driver records a file it cannot READ as an error rather than skipping it; a non-empty error list makes the compiler fail, so `bl.cli/run-argv` is never reached. The output names the offending file but never says the command did not run — it reads like the command failed.

This is the same law this report established for `env.bl` — **a broken file degrades, never stops a command** — violated one layer down, where it costs more, because it takes the warm session's own daemon with it. Observed live while a second session was editing this repository (13 unreadable sources).

Filing it rather than patching it: the honest fix narrows what the Mix compiler treats as fatal (an ordinary source that cannot be read should be skipped, and its dependents fail on their `require` — which is what the build driver already does and what its own comment argues for), and that is a change to global build semantics and CI gates. FUP-050 carries the mechanism, the reproduction and the options.

The red test that turned out to be a bug

One case kept failing only inside the full suite — a `bl test` run inside a long-running VM — while passing four times out of four on its own. Passing standalone is not evidence of correct; it is evidence that the trigger lay elsewhere.

Printing the state AT THE FAILURE SITE, instead of guessing at suite order, gave the answer in one run:

```
$ the diagnostic, printed by the failing case in a real suite run
```

```
DIAG exit=1 is-report/4=false guard=true beam=.../ebin/Elixir.BeamLisp.Ns.Core.beam
```

The assertion runtime was MISSING FROM THE COMPILED CORE MODULE while the check that guards it said everything was fine. `ward`'s `deftest-available?` asked only whether `deftest` resolves in core's ENV — but a compiled test file does not call an env var: `is` expands into a call to `BeamLisp.Ns.Core."is-report"/4`, and `deftest` into `test-record/6`. Those live in the MODULE, and a later reload of `core` puts the module back from its beam (which carries `core.bl` alone — the test library is not in it) while core's env keeps the macro's home. The guard said “available”, `ensure-test-lib!` skipped the load, and every file in the run died on a function the guard had just certified.

DECISION

The guard now requires the env AND the compiled module (`assertion-runtime-loaded?` — `function_exported/3` for `is-report/4` and `test-record/6`).

Because — the env is where the macro LIVES; the module is what a compiled file CALLS. A check that sees only one of them certifies a library that cannot run.

Reproduced deterministically — purge core, load it back from its beam, run `bl test` in-VM — and that reproduction is now a regression test rather than a suite-order mystery:

```
$ mix test test/beam_lisp/test_verb_test.exs
```

```
8 passed
```

```
including `a test library that vanished from core's MODULE is re-loaded`
```


The suite went from 1535–1541 of 1541 (this class failing intermittently) to **1542 passed, 0 failures** — and faster, 234s against 421s, because every file had been paying for a library that was never loaded.

Two bugs the machine found

The last pass of this work was not new surface. It was the two defects the workstation exposed once this tree came back up — both in code this plan ships, both reproduced before and after the fix, both now covered by a test that fails without it.

The page that waited for the file server

The dashboard stopped answering: `GET /` hung, first in ExUnit at the 60-second test timeout, then at 120 seconds to a raw `:gen_tcp` client, with the daemon idle and its `:ui` port bound. Reproduced in a bare `mix run`, so it was never the test harness. The stalled process names its own blocker:

```
Process.info(stalled, :monitors)    #> [process: #PID<0.53.0>]
Process.whereis(:file_server_2)    #> #PID<0.53.0>
```

Stack, read from the process rather than guessed: `Gateway.url/2` → `Paths.runtime_dir/0` → `ensure_secure_dir/1` → `:file.call/2` → `gen_server:call(:file_server_2, ...)` — with `infinity` as the timeout. This host runs every Elixir VM with one dirty-IO scheduler:

```
ELIXIR_ERL_OPTIONS=+S 4:4 +SDcpu 2:2 +SDio 1:1 +sbwt none +sbwtdio none
```

so a file-server round trip queues behind any other dirty-IO operation in the same VM and waits without a deadline. The page was paying for that queue once per element: `Gateway.url/1` looks the port up per call, and the lookup reads the runtime dir and the endpoint file — 2N file round trips to draw a page with N named ports.

- Fix: the model asks the gateway for its port ONCE per render (one file round trip, not 2N) and hands the number to `Gateway.url/2`.
- Observed: `GET /` → 200 in 30 ms, 764 file-server reductions, 4836-byte body (was: no answer in 120 s).
- The host fact is recorded where host facts belong (`~/.agents/AGENTS.md`), with the diagnosis recipe, because it will bite any Elixir project here.

A sibling tree switched this tree's page off

Once the page answered, its test still failed — for an unrelated reason that turned out to be the sharper bug. This tree's daemon printed:

```
ui: not served - {:taken, %{name: "ui", port: 51981, root: ".../beam-lisp--autovcs"}}
```

A **different checkout's** session held the name `ui`, and the registry — which keys claims by name alone, on purpose, so that two trees cannot both promise the same NUMBER — refused the second tree its own ephemeral port. The user sees a session with no page, for a reason they can neither see nor fix.

The key now says what kind of claim it is:

- A CHOSEN port (`:ports {:ui 7700}`) is a promise about a NUMBER. The name is the key, machine-wide: a second tree is refused, naming the owner.
- An EPHEMERAL port is nobody's promise — the OS picked the number — so the claim is the SESSION's address and is keyed by name AND tree.

- Observed after: `ui: http://beam-lisp.test → 127.0.0.1:52435 (ephemeral — while pin it in env.bl with :ports {:ui 7700})`
the sibling tree still holds `ui`.
- Regression test on both sides: “two trees may each hold the same ephemeral name” (ExUnit and the `bl.ports` corpus), plus “a chosen port is machine-wide” for the refusal that must still happen.

What the docs had to say about it

The walkthrough gate caught the behaviour change honestly: after the keying fix,

```
$ bl doc run docs/dev/the-warm-session.bl.md --check
```

stored results are stale; run `bl doc run` to refresh

and the refresh showed the prose was now wrong in two cells: a release that did not name the tree left its claim behind (`(true false) → (true true)`), and the “two trees are refused” example — which is only true of a **chosen** port — died on an **argument error** because the second claim now succeeds. Both cells were rewritten and the section now states the rule instead of the old behaviour: every checkout gets its own session page at once; a chosen port is the promise the machine keeps once.

```
$ bl doc run docs/dev/the-warm-session.bl.md
```

6 cells, 0 cell errors

What is not done

-
- G1, above — the dashboard is server-rendered: one read-model, no keyed diffs, no convergence between tabs.
 - G2, above — `u/kw` silently drops what it cannot express.
 - G3, above — one mid-write source file stops every `bl` command, the daemon included. FUP-050 has the mechanism and the reproduction.
 - The verb livebooks `bl.env`, `bl.test`, `bl.ports` are gated clean (their definition cells are `silent`, so they run without storing function references as “results”). Every OTHER `priv/std/bl/*.bl.md` still reports stale for the same reason — FUP-054 has the mechanism and the confirmed mechanical fix. Running `bl doc run` on them is NOT the fix.
 - A file read in a session can still stall behind another operation in the same VM — this host gives every Elixir VM one dirty-IO scheduler. The page now asks once per render instead of once per port; the rest is the host’s setting, not a property of the session.
 - The dashboard REPL pane (FUP-038) and the editing/file-browsing spike (FUP-039) remain follow-ups, as planned.
 - The plan listed five walkthroughs (`env.bl`, `daemon`, `tasks`, `ports`, `ui`); two shipped, covering the same ground in one pass each. The verb livebooks (`bl.env`, `bl.test`, `bl.ports`) are the other half of that documentation.

The red corpus, attributed

The corpus (`bl test test/bl/`) is red as this report is written, and the attribution matters more than the count:

```
$ mix bl test test/bl/
```

```
147 file(s) passed, 30 failed, 1 incoherent (one run; the counts move with their edits)
```

Every failing file is in the surface the **other** session is editing in this same checkout at the same time — the solver and its callers (reporting `smt/emit: nil is not an SMT term`, `:verdict :unknown`, and one ETS table identifier does not refer to an existing ETS table

- `reload.migrate-test`, `reload.upgrade-test`, `reload.verify-adversarial-test` — the SMT-backed verification of a migration.
- `system.anf-smt-test`, `system.smt-defn-test`, `system.smt-fragment-test`, `system.smt-quot-rem-test`, `system.theories-test`, `system.repair-test`, `system.seam-test`, `system.interproc-wins-test` — the solver layer.
- `veritas-test` and `veritas.*` (fault, mock, covers-symbolic, hypothesis-symbolic, theories, tuple-positional) — 10 of 14 red, their largest single file.
- `system.lsp-test`, `system.lsp-cli-test`, `system.linear-test`, `system.check-test`, `system.mcp-project-mount-test`, `system.system-test` — files they were mid-writing (`priv/lib/lsp*`, `priv/lib/system/linear.bl`, `priv/lib/mcp/tools.bl`).
- Not one failure text mentions this work's surfaces. `test/bl/smt_test.bl` itself is green (28 passed) while the layers above it are not.

That last point is the one worth keeping: the warm runner stayed green for this work while the tree around it was broken, because a verb loads by name and the broken files were in namespaces this work does not touch.

The shape of the thing

BEFORE

Twenty-one verbs, each finding a project's files its own way; a daemon and an isolated test runner that did not know about each other; warmth and isolation as commands a developer had to choose between.

AFTER

One project file the tooling reads about itself; one runner whose isolation is the default; one session per tree with one address; one read-model that the terminal and the browser both render. Warmth is an optimization, isolation is a property, and the CLI is a remote control.